DIVISION ALGORITHMS

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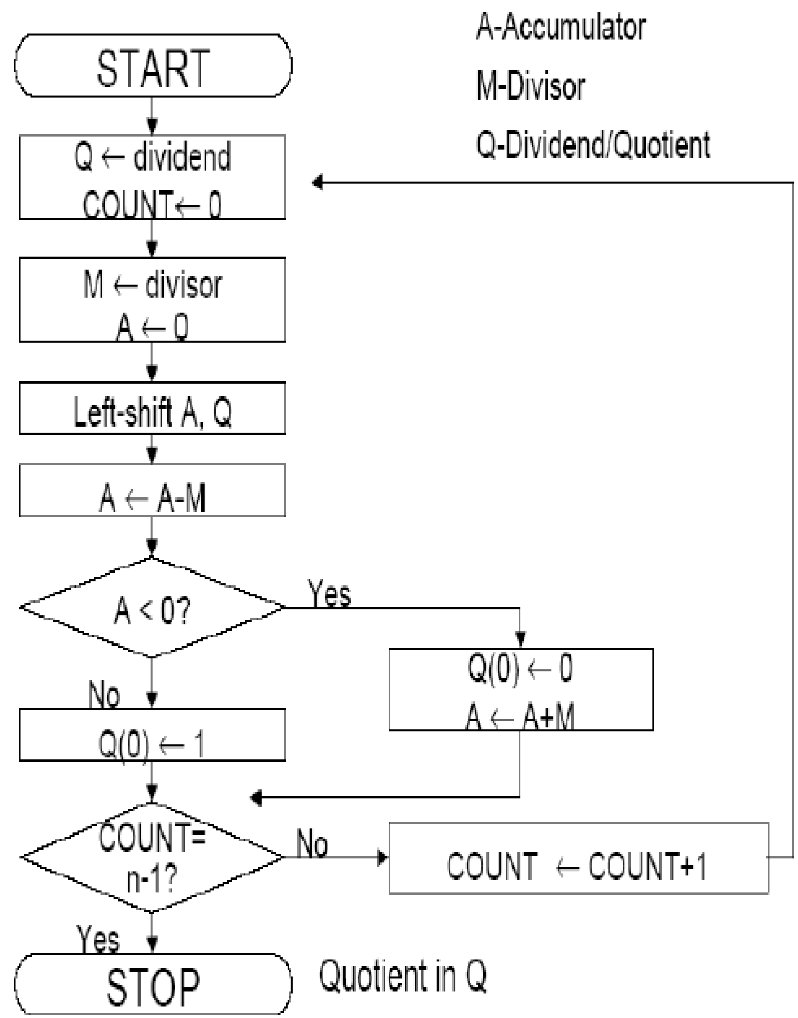
Restoring Division

- Input:
  - M – positive divisor (n-bit)
  - Q – positive dividend (n-bit)
- Output:
  - Q – Quotient
  - A – Remainder
- Begin
  - A is set to 0.
  - Shift A and Q left one binary position
  - A ← A – M
  - If sign of A is 1
    - q_0 ← 0 and A ← A + M (Restore A)
  - Else
    - q_0 ← 1
- End



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	A	Q	SC
	0000	1000	4
LSH AQ	0001+	0000	
Subtract A-M	1101		
	1110+	0000	
Restore A	0011		
	0001		3
LSH AQ	0010+	0000	
Subtract A-M	1101		
	1111+	0000	
Restore A	0011		
	0010		2
LSH AQ	0100+	0000	
Subtract A-M	1101		
	0001	0001	1
LSH AQ	0010+	0010	
Subtract A-M	1101		
	1111+	0010	
Restore A	0011		
	0010		0

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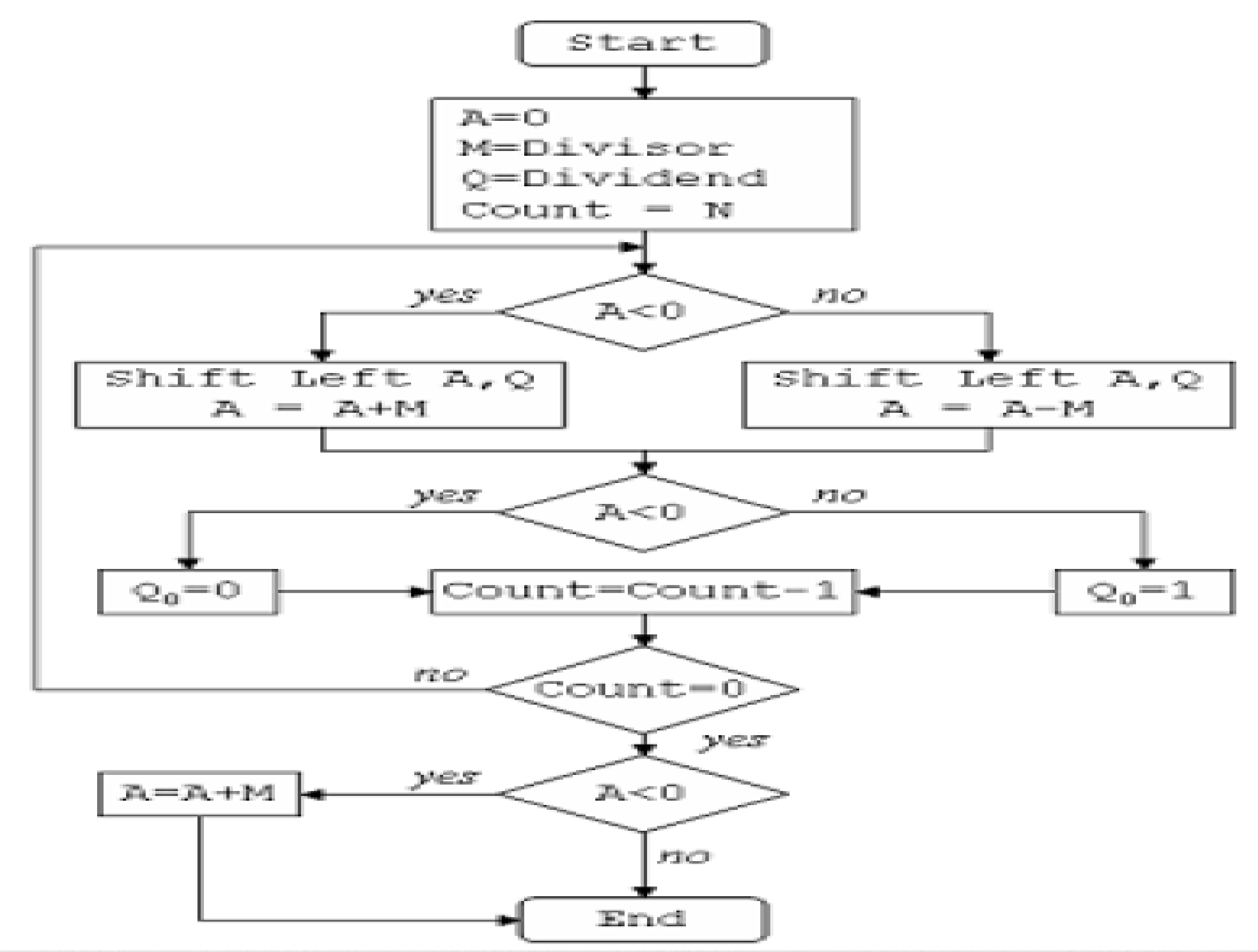
A	Q	M = 0011	
0000	0111	Initial values	
0000	1110	Shift	1
1101		A = A - M	
0000	1110	A = A + M	
0001	1100	Shift	2
1110		A = A - M	
0001	1100	A = A + M	
0011	1000	Shift	3
0000		A = A - M	
0000	1001	$Q_0 = 1$	
0001	0010	Shift	4
1110		A = A - M	
0001	0010	A = A + M	

Non-Restoring Division

- Input:
  - M – positive divisor (n-bit)
  - Q – positive dividend (n-bit)
- Output:
  - Q – Quotient
  - A – Remainder
- Begin
  - $A \leftarrow 0$
  - Do n times
    - If the sign of A is 0
      - Shift A and Q left one bit position and $A \leftarrow A - M$
    - else
      - Shift A and Q left one bit position and $A \leftarrow A + M$
    - If Sign of A is 0
      - $q_0 \leftarrow 1$
    - Else
      - $q_0 \leftarrow 0$
  - If sign of A is 1
    - $A \leftarrow A + M$
- End



Start
A=0
M=Divisor
Q=Dividend
Count = N
A<0
yes
no
Shift Left A,Q
A = A+M
Shift Left A,Q
A = A−M
A<0
yes
no
Q0=0
Count=Count−1
Q0=1
Count=0
no
yes
A<0
yes
A=A+M
no
End

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	A	Q	SC
	0000	1000	4
LSH AQ	0001+	0000	
Subtract A-M	1101		
	1110	0000	3
LSH AQ	1100+	0000	
Add A+M	0011		
	1111	0000	2
LSH AQ	1110+	0000	
Add A+M	0011		
	0001	0001	1
LSH AQ	0010+	0010	
Subtract A-M	1101		
	1111+	0010	0
Restore A	0011		
	0010		